

作品 1：三層 IP 網路架構 (Routed-Access) 與 Rocky Linux 及 ISO 27001:2022 安全規範整合實作

平台：Cisco IOS/ GNS3 / Rocky Linux

安全框架：27001:2022

關鍵技術：L3 Routed Access, EtherChannel, Syslog, DHCP, HSRP, PAT, NTP, OSPF, Stratis File System (auto mount), ISO 27001:2022 Control Implementation

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商業背景概述：

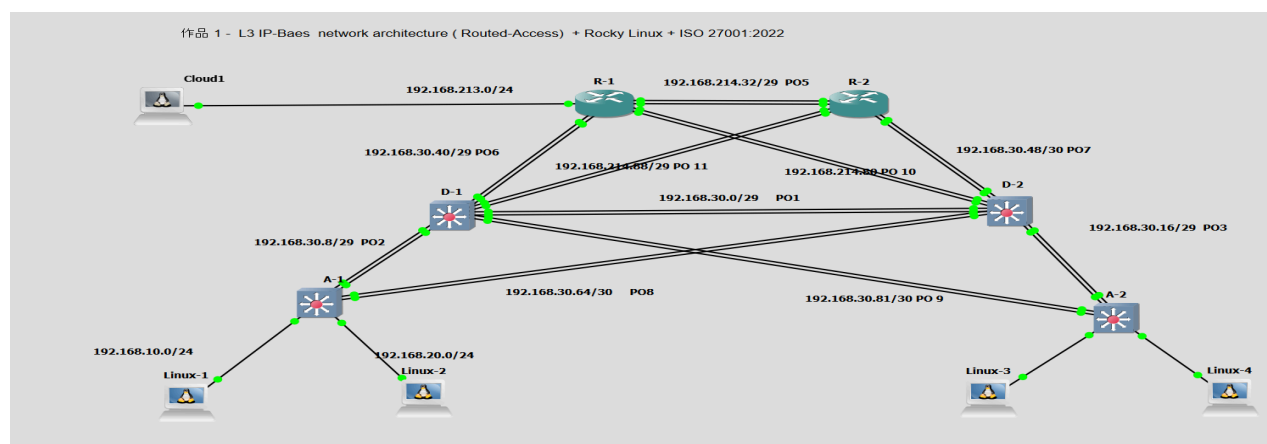
一家區域性金融服務機構正在升級其資料中心基礎設施，以符合ISO 27001:2022 認證要求，同時更換老舊且容易發生故障的二層交換網路。先前的架構存在諸多問題，例如：中斷期間生成樹協定 (STP) 收斂速度慢、頻寬可擴展性有限以及安全控制分散。。

為了消除單點故障、滿足審計合規性要求並支援機架的快速擴展，該機構部署了路由存取三層IP架構，並整合了加固型Rocky Linux中央管理主機

系統架構與合規性概述：

- 高可用性與擴充性 (Availability & Scalability)：全網採用全三層路由 (Routed Core & Access) 結合 Port-Channel 多線路冗餘，適合在要求零中斷服務企業環境中。此架構採用 Routed Access (L3 路由到接入層)，透過 OSPF 搭配 ECMP實現毫秒級收斂與流量負載分流，未來新增機架 (Rack) 或樓層，只需分配新的 IP Subnet 並發布路由，不影響原有網路拓撲。配合 Rocky Linux 建立集中式服務 (Syslog 遠端日誌伺服器、NTP 時間同步伺服器、DHCP) 與 Stratis 存取，能有效降低營運維護成本並符合系統 hardening (強化安全) 要求。
- ISO 27001:2022 安全框架控制 (Security Controls)：「ISO 27001:2022 安全控制措施 (Security Controls)：A.5.17 身分驗證資訊 (強制執行使用者密碼生命週期管理與定期輪調、A.8.2 特權存取權 (防止未授權的 Root 權限存取)、A.8.5 安全性認證 (取消 Root 外部直接登入管理，配備 SSHv2 存取權、A.8.15日誌留存 (Rocky Linux與L3設備集中系統日誌審計)、A.8.17時脈同步 (確保整體網路架構與伺服器環境系統時脈一致)、A.8.20網路安全性 (僅保留必要開放端口，並包圍不必要的網路服務與高風險協議) 及A.8.24密碼學的使用 (保護網路傳輸與管理界面之資訊機密性)，以確保整體架構符合國際資安標準。

拓撲結構：



IP Address Table

Device	Port	IP	Mask	Description
R1	g1	192.168.213.12	/24	connect to Gateway
	g2	shut down		
	g3	po5		connect to R-2
	g4	po5		connect to R-2
	g5	po6		connect to D-1
	g6	po6		connect to D-1
	g7	po6		connect to D-2
	g8	po6		connect to D-2
	po 5	192.168.30.33	/29	connect to R-2
	po 6	192.168.30.42	/29	connect to D-1
	po 10	192.168.30.81	/29	connect to D-2
	standby 5	192.168.30.35		connect to R-2 (HSRPv2)
	standby 6	192.168.30.46		connect to D-1 (HSRPv2)
	standby 10	192.168.30.86		connect to D-2 (HSRPv2)
D-2	g0/0	shut down		
	g0/1	shut down		
	g0/2	po1		connect to D-1
	g0/3	po1		connect to D-1
	g1/0	po3		connect to A-2
	g1/1	po3		connect to A-2
	g1/2	po7		connect to R-2
	g1/3	po7		connect to R-2
	g2/0	po8		connect to A-1
	g2/1	po8		connect to A-1
	g2/2	po10		connect to R-1
	g2/3	po10		connect to R-1
	g3/0	shut down		
	g3/1	shut down		
	g3/2	shut down		
	g3/3	shut down		
	po 1	192.168.30.2	/29	connect to D-1
	po 3	192.168.30.18	/29	connect to A-2
	po 7	192.168.30.42	/29	connect to R-2
	po 8	192.168.30.52	/29	connect to A-1
	po 10	192.168.30.82	/29	connect to R-1
	standby 1	192.168.30.6		connect to D-1 (HSRPv2)
	standby 3	192.168.30.22		connect to A-2 (HSRPv2)
	standby 7	192.168.30.46		connect to R-2 (HSRPv2)
	standby 8	192.168.30.52		connect to A-1 (HSRPv2)
	standby 10	192.168.30.86		connect to R-1 (HSRPv2)

驗證 PAT 配置

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
tcp  192.168.213.12:5068 192.168.10.100:33732 34.49.197.38:443   34.49.197.38:443
tcp  192.168.213.12:5067 192.168.10.100:33722 34.49.197.38:443   34.49.197.38:443
tcp  192.168.213.12:5062 192.168.20.110:56328 34.49.197.38:443   34.49.197.38:443
tcp  192.168.213.12:5069 192.168.10.100:33742 34.49.197.38:443   34.49.197.38:443
Total number of translations: 4
```

驗證 DHCP 地址釋出 Linux-1 and Linux-2

- From Rocky Linux (DHCP server - IP release to Linux-1 & Linux-2)

```
Aug  8 10:39:42 localhost kea-dhcp4[4967]: 2026-08-08 10:39:42.696 INFO [kea-dhcp4.leases
/4967.140352074102464] DHCP4_LEASE_REUSE [hwtype=1 0c:ca:02:c8:00:00], cid=[01:0c:ca:02:c8
:00:00], tid=0x19684f9e: lease 192.168.20.110 has been reused for 6727 seconds
Aug  8 10:49:02 localhost kea-dhcp4[4967]: 2026-08-08 10:49:02.031 INFO [kea-dhcp4.leases
/4967.140352057317056] DHCP4_LEASE_ALLOC [hwtype=1 0c:e9:b4:e4:00:00], cid=[01:0c:e9:b4:e4
:00:00], tid=0x6926f326: lease 192.168.10.100 has been allocated for 7200 seconds
```

```
[root@rocky-cloud ~]# nmcli dev show | grep IP4
IP4.ADDRESS[1]: 192.168.10.100/24
IP4.GATEWAY: 192.168.10.1
IP4.ROUTE[1]: Linux-1 obtain DHCP IP from Server dst = 192.168.10.0/24
IP4.ROUTE[2]: dst = 0.0.0.0/0, nh
IP4.DNS[1]: 8.8.8.8
IP4.ADDRESS[1]: 127.0.0.1/8
IP4.GATEWAY: --
[root@rocky-cloud ~]#
```

```
[root@rocky-cloud ~]# nmcli dev show | grep IP4
IP4.ADDRESS[1]: 192.168.20.110/24
IP4.GATEWAY: Linux-2 obtain 192.168.20.1
IP4.ROUTE[1]: DHCP IP from dst = 192.168.20.0/24, nh = 0.0.0.0, mt = 100
IP4.ROUTE[2]: Server dst = 0.0.0.0/0, nh = 192.168.20.1, mt = 100
IP4.DNS[1]: 8.8.8.8
IP4.ADDRESS[1]: 127.0.0.1/8
```

驗證 HSRP 備用路由器

```
D-2#sh standby br
P indicates configured to preempt.
|
Interface    Grp  Pri P State    Active      Standby      Virtual IP
Po1          1    100   Standby 192.168.30.1 local        192.168.30.6
Po3          3    110 P Active  local       192.168.30.18 192.168.30.22
Po7          7    100   Standby 192.168.30.41 local        192.168.30.46
Po8          8    110 P Active  local       192.168.30.50 192.168.30.52
Po10         10   100   Standby 192.168.30.81 local        192.168.30.86
```

```
.
interface Port-channel3
 no switchport
 ip address 192.168.30.19 255.255.255.248
 standby version 2
 standby 3 ip 192.168.30.22
 standby 3 priority 110
 standby 3 preempt
 ip ospf authentication key-chain OSPF-KEY
 ip ospf 100 area 0
```

驗證 OSPF-Neighborship with Authentication

```
D-2#sh ip os neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
192.168.213.12  1     FULL/DR         00:00:38   192.168.30.81  Port-channel10
192.168.30.89  1     FULL/DR         00:00:38   192.168.30.41  Port-channel7
192.168.30.75  1     FULL/BDR        00:00:35   192.168.30.18  Port-channel3
192.168.30.90  1     FULL/DR         00:00:38   192.168.30.1   Port-channel1
D-2#
```

```
key chain OSPF-KEY
 key 200
   key-string 7 0110050A5A
   cryptographic-algorithm hmac-sha-256
 password encryption aes
```

驗證集中式日誌記錄 (Centralize Log Server)

```
root@localhost:/var/log/remote# ls
127.0.0.1      192.168.30.1  192.168.30.18  192.168.30.75
192.168.213.12 192.168.30.11 192.168.30.19
root@localhost:/var/log/remote# cd 192.168.30.1
root@localhost:/var/log/remote/192.168.30.1# ls
2026-08-08.log 2026-08-09.log
root@localhost:/var/log/remote/192.168.30.1# cat 2026-08-08.log
Aug  8 20:12:37 192.168.30.1 139: *Aug  8 10:05:36.050: %OSPF-5-ADJCHG: Process 100, Nbr 1
92.168.215.1 on Port-channel11 from LOADING to FULL, Loading Done
root@localhost:/var/log/remote/192.168.30.1# cd ..
root@localhost:/var/log/remote# cd 192.168.30.19
root@localhost:/var/log/remote/192.168.30.19# ls
2026-08-08.log
root@localhost:/var/log/remote/192.168.30.19# cat 2026-08-08.log
Aug  8 18:46:29 192.168.30.19 103: *Aug  8 09:56:00.424: %GRUB-5-CONFIG_WRITING: GRUB conf
iguration is being updated on disk. Please wait...
Aug  8 18:46:29 192.168.30.19 104: *Aug  8 09:56:01.121: %GRUB-5-CONFIG_WRITTEN: GRUB conf
iguration was written to disk successfully.
```

```
A-2(config)#logg
A-2(config)#logging on
A-2(config)#logging host 192.168.213.14
A-2(config)#logging h
*Aug  8 06:07:13.164: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.213.14 port 514 started - CLI initiate
% Incomplete command.

A-2(config)#logging tr
A-2(config)#logging trap 5
```

驗證 NTP和伺服器時間同步

```
#
# add network in /etc/chrony.conf
allow 192.168.213.0/24
allow 192.168.30.0/24
```

```
root@localhost:~# chronyc sources -v
```

```

.-- Source mode  '^' = server, '=' = peer, '#' = local clock.
/  .-- Source state '*' = current best, '+' = combined, '-' = not combined,
| /      'x' = may be in error, '~' = too variable, '?' = unusable.
||
||          Reachability register (octal) --.          |      .- xxxx [ yyyy ] +/- zzzz
||          Log2(Polling interval) --.          |      |      xxxx = adjusted offset,
||                                  \          |      |      yyyy = measured offset,
||                                  |          |      |      zzzz = estimated error.
||                                  |          |      \
||
MS Name/IP address             Stratum Poll Reach LastRx Last sample
=====
^+ 103.186.118.217                2    6    17    32    +593us[ +593us] +/- 8188us
^- 123-204-232-128.adsl.dyn>      2    6    17    32    +1365us[+1365us] +/-  56ms
^- 103.186.118.222                2    6    17    32     -475us[ -475us] +/-  13ms
^* 103.186.118.212                2    6    17    32     -172us[-1553us] +/- 9388us
```

```
D-1#sh ntp associations
```

```

address          ref clock      st   when   poll reach  delay  offset  disp
*~192.168.213.14  103.186.118.212 3    22    64     3   7.205 4761.23 1937.8
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

驗證Stratis File System : auto mount to client

```
root@localhost:~# stratis filesystem list
```

Pool	Filesystem	Total / Used / Free / Limit	Device	UUID
pool1	fs1	1 TiB / 546 MiB / 1023.47 GiB / None	/dev/stratis/pool1/fs1	3dac3735-2032-4702-b249-32cca0293a46
pool1	snapshot1	1 TiB / 546 MiB / 1023.47 GiB / None	/dev/stratis/pool1/snapshot1	41202873-6a39-474a-802a-ed24550098dd

```
vda4 252.74 0 0.00 0 part /
```

```
[root@rocky-cloud ~]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/vda4	8.4G	1.1G	7.4G	13%	/
devtmpfs	594M	0	594M	0%	/dev
tmpfs	633M	0	633M	0%	/dev/shm
tmpfs	254M	552K	253M	1%	/run
tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
/dev/vda3	936M	279M	658M	30%	/boot
/dev/vda2	200M	14M	186M	7%	/boot/efi
tmpfs	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	1.0M	0	1.0M	0%	/run/credentials/serial-getty@ttyS0.service
tmpfs	127M	4.0K	127M	1%	/run/user/1000
192.168.213.14:/srv/stratis_share	17G	5.4G	12G	32%	/mnt/client_stratis

遵守 ISO 27001:2022 - A.8.2—特權存取權限 (要求對特權存取權限的分配和使用進行嚴格控制和限制), A.8.5—安全認證 (緩解針對管理員帳戶的暴力破解攻擊)

```
root@localhost:~# grep -i "PermitRootLogin" /etc/ssh/sshd_config
```

```
#PermitRootLogin prohibit-password
```

```
# the setting of "PermitRootLogin prohibit-password".
```

```
PermitRootLogin no
```

遵守 ISO 27001:2022 : A.8.20—網路安全 (最少的開放網路端口), A.8.9— 配置管理 (服務最小化)

```
root@localhost:~# firewall-cmd --list-services
```

```
cockpit dhcp dhcpv6-client mounsd nfs ntp rpc-bind ssh
```

遵守 ISO 27001:2022 : A.5.17—身份驗證訊息 (強制執行使用者密碼生命週期管理和定期輪調), A.8.9 配置管理(強制執行標準化的 Linux 帳戶老化和密碼原則基準)

```
root@localhost:~# chage --maxdays 60 peter
```

```
root@localhost:~# chage --list peter
```

Last password change	: Aug 09, 2026
Password expires	: Oct 08, 2026
Password inactive	: never
Account expires	: never
Minimum number of days between password change	: 0
Maximum number of days between password change	: 60
Number of days of warning before password expires	: 7

遵守 ISO 27001:2022 : A.8.15—日誌記錄 (產生、匯總和儲存遠端事件日誌，以便進行分析和稽核)

```
root@localhost:/var/log/remote# ls
127.0.0.1 192.168.213.12 192.168.30.1 192.168.30.11 192.168.30.18 192.168.30.19 192.168.30.75
root@localhost:/var/log/remote# cat 192.168.30.11
cat: 192.168.30.11: Is a directory
root@localhost:/var/log/remote# cd 192.168.30.11
root@localhost:/var/log/remote/192.168.30.11# ls
2026-08-08.log 2026-08-09.log
root@localhost:/var/log/remote/192.168.30.11# cat 2026-08-09.log
Aug  9 14:11:39 192.168.30.11 93: *Aug  9 06:11:25.889: %HSRP-5-STATECHANGE: Port-channel8 Grp 8 state Sp
eak -> Standby
Aug  9 14:12:01 192.168.30.11 94: *Aug  9 06:11:47.447: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vi
os_l2-adventerprisek9-m' passed code signing verification
Aug  9 15:41:42 192.168.30.11 98: *Aug  9 07:40:17.045: %SYS-5-CONFIG_I: Configured from console by conso
le
Aug  9 15:44:25 192.168.30.11 101: *Aug  9 07:43:00.584: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was w
ritten to disk successfully.
root@localhost:/var/log/remote/192.168.30.11# S
```

遵守 ISO 27001:2022 : A.8.17—時鐘同步 (確保整個環境中的系統時鐘與標準時間源保持一致)

```
R-2#sh ntp associations
```

address	ref clock	st	when	poll	reach	delay	offset	disp
*~192.168.213.14	103.186.118.215	3	259	256	377	1.799	1116404	3.329
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured								

```
D-1#sh ntp associations
```

address	ref clock	st	when	poll	reach	delay	offset	disp
*~192.168.213.14	103.186.118.212	3	22	64	3	7.205	4761.23	1937.8
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured								

遵守 ISO 27001:2022 : A.8.20 網路安全 (關閉阻止不需要的網路服務)

```
no cdp run
no ip http server
no ip http secure-server
no ip source-route
```


遵守 27001:2022 - A.8.24 密碼學的使用 (有效地使用加密技術進行控制，以保護資訊的機密性、真實性和完整性。)

```
no ipv6 cef
!
key chain OSPF-KEY
  key 200
    key-string 7 110A1A0B16
    cryptographic-algorithm hmac-sha-256
password encryption aes
```

```

R-2#sh run | s line
line con 0
  stopbits 1
line vty 0 4
  password 7 022527753B
  login
  transport input ssh
line vty 5 15
  password 7 022527753B
  login
  transport input ssh
```